

Modelling
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Start a New Modelling Project

Target Sequence(s):
(Format must be FASTA, Clustal, plain string, or a valid UniProtKB AC)

Target: 145
Target: 290
Target: 302

Add Hetero Target
Reset

Project Title:
Email:

Search For Templates
Build Model

By using the SWISS-MODEL server, you agree to comply with the following [terms of use](#) and to cite the corresponding [articles](#).

You are currently not logged in - to take advantage of the workspace, please [log in](#) or [create an account](#).

(There is no requirement to create an account to use any part of SWISS-MODEL, however you will gain the benefit of seeing a list of your previous modelling projects [here](#).)

把序列输入到 SWISS-MODEL，寻找相似可替代序列，找到相似蛋白 H6BDW6.1.A 建模如下：

Summary
Templates **50**
Models
Project Data

Template Results

Templates
Quaternary Structure
Sequence Similarity
Alignment
More

Sort	Coverage	GMQE	QSQE	Identity	Method	Oligo State	Ligands
 H6BDW6.1.A UniProtKB entry unknown, most likely obsolete AlphaFold DB model of H6BDW6 (gene: unknown, organism: unknown)		0.95	-	81.46	AlphaFold v2	monomer ✓	None
	Method AlphaFold v2 Found By AFDB search Seq Similarity 0.55 Biounit Oligo State monomer Target Prediction It is only possible to build a monomer.	Build Model					
Target: MSPOLQNTITFMKSRPSSSRVSTREPPAAEQIATVLPYAGDVERGKASANGYGEYFPFGVAGDITVYVHGGGYMGSYNT 85 H6BDW6.1.A: MSPOLQNTITFMKSRPSSSRVSTREPPAAEQIATVLPYAGDVERGKASANGYGEYFPFGVAGDITVYVHGGGYMGSYNT 85 Target: TANTIAQIAAACARAFIDYRLAPENPPAGLDVAGYRVLLGGYDPAFLYIGGDSAGGGLSATLQRLREAGNPLPAATVLL 170 H6BDW6.1.A: TANTIAQIAAACARAFIDYRLAPENPPAGLDVAGYRVLLGGYDPAFLYIGGDSAGGGLSATLQRLREAGNPLPAATVLL 170 Target: LSPFTDLDTIGESIKTRRDADPSIDPARLPDFAREYHPDGLRPLYSPLVADFGLPFPMITQVDAEYLLDSTRYTEREAGQDYLEYNDEMYFQWAFNPLPAVAATERIGER 235 H6BDW6.1.A: LSPFTDLDTIGESIKTRRDADPSIDPARLPDFAREYHPDGLRPLYSPLVADFGLPFPMITQVDAEYLLDSTRYTEREAGQDYLEYNDEMYFQWAFNPLPAVAATERIGER 235 Target: MSHGGASAAA 302 H6BDW6.1.A: MSHGGASAAA 302							
☐ 7at2.1.A EstD11 S144A Crystal structure of inactive EstD11 S144A		0.85	-	52.36	X-ray, 1.4Å	monomer ✓	None
☐ 5gms.1A Esterase Crystal structure of the mutant S202W/I203F of the esterase E40							

Build Models 1

Clear Selection

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AF-H6BDW6-F1-model_v4

Currently SWISS-MODEL includes models from AlphaFold Database version 4. The latest AlphaFold Database version is 6. The coordinates on this page are the ones currently used in the SWISS-MODEL pipeline.

The latest AlphaFold Database version 6 models will not be significantly different than version 4, however UniProtKB content has changed a lot since version 2021_04 which may result in changes in sequence, or the entry no longer found in AlphaFold Database v6.

Coordinates
PDB Format
Compare

Method
AlphaFold v2.0 (model_2_ptm)

Links
The following links will show the latest AlphaFold Database version 6 model, if available, for this UniProtKB accession.

SWISS-MODEL Repository
AlphaFold Database (v6)

A: 45
A: 90
A: 135
A: 180
A: 225
A: 270
A: 302

参考代码如下：

```
#install.packages("yulab.utils")
library(yulab.utils)

## 检索 NCBI 数据库，获取序列信息
pload(rentrez)

x = entrez_fetch(
  db = "protein",
  id = c("AUF80930", "AUF80945"),
  rettype = "fasta"
)

cat(x)
## 蛋白质结构可视化
pload(r3dmol)

auf80930 <- readLines("AUF80930.pdb")#可以从 PDB 上面下载、替换各种蛋白质
auf80945 <- readLines("AUF80945.pdb")

r3dmol() %>%
  m_add_model(data = auf80945, format = 'pdb') %>%
  m_zoom_to() %>%#输出变输入
  m_set_style(style = m_style_cartoon(color = 'spectrum')) %>%
  m_add_style(
    style = m_style_stick(),
    sel = m_sel(resi = c(44, 238, 269), chain = 'A')
  )

r3dmol() %>%
  m_add_model(data = auf80930, format = 'pdb') %>%
  m_zoom_to() %>%
  m_set_style(style = m_style_cartoon(color = 'spectrum')) %>%
  m_add_style(
    style = m_style_stick(),
    sel = m_sel(resi = c(144, 238, 269), chain = 'A')
  ) %>%
  m_add_style(
    style = m_style_sphere(),
    sel = m_sel(resi = 199, chain = 'A')
  ) %>%
  m_add_label(text = "Ser144", sel = m_sel(resi = 144)) %>%
  m_add_label(text = "Glu238", sel = m_sel(resi = 238)) %>%
  m_add_label(text = "His269", sel = m_sel(resi = 269)) %>%
  m_add_label(text = "Leu199", sel = m_sel(resi = 199)) %>%
  m_add_surface(style = m_style_surface(opacity = 0.4)) %>%
  m_zoom_to() %>%
```

```

m_spin()

y = r3dmol() %>%
  m_add_model(data = auf80945, format = 'pdb') %>%
  m_zoom_to() %>%
  m_set_style(style = m_style_cartoon(color = 'spectrum')) %>%
  m_add_style(
    style = m_style_stick(),
    sel = m_sel(resi = c(144, 238, 269), chain = 'A')
  ) %>%
  m_add_style(
    style = m_style_sphere(),
    sel = m_sel(resi = 199, chain = 'A')
  ) %>%
  m_add_label(text = "Ser144", sel = m_sel(resi = 144)) %>%
  m_add_label(text = "Glu238", sel = m_sel(resi = 238)) %>%
  m_add_label(text = "His269", sel = m_sel(resi = 269)) %>%
  m_add_label(text = "Leu199", sel = m_sel(resi = 199)) %>%
  m_add_surface(style = m_style_surface(opacity = 0.4)) %>%
  m_zoom_to()
#####
dna<-readLines("7AT3.pdb")
r3dmol() %>%
  m_add_model(data = dna, format = 'pdb') %>%#练习，只需要改 data 输入的名称
  m_zoom_to() %>%#输出变输入
  m_set_style(style = m_style_cartoon(color = 'spectrum')) %>%
  m_add_style(
    style = m_style_stick(),
    sel = m_sel(resi = c(44, 238, 269), chain = 'A')
  )
r3dmol() %>%
  m_add_model(data = dna, format = 'pdb') %>%
  m_zoom_to() %>%
  m_set_style(style = m_style_cartoon(color = 'spectrum')) %>%
  m_add_style(
    style = m_style_stick(),
    sel = m_sel(resi = c(144, 238, 269), chain = 'A')
  ) %>%
  m_add_style(
    style = m_style_sphere(),
    sel = m_sel(resi = 199, chain = 'A')
  ) %>%
  m_add_label(text = "Ser144", sel = m_sel(resi = 144)) %>%
  m_add_label(text = "Glu238", sel = m_sel(resi = 238)) %>%
  m_add_label(text = "His269", sel = m_sel(resi = 269)) %>%
  m_add_label(text = "Leu199", sel = m_sel(resi = 199)) %>%
  m_add_surface(style = m_style_surface(opacity = 0.4)) %>%
  m_zoom_to() %>%

```

m_spin()